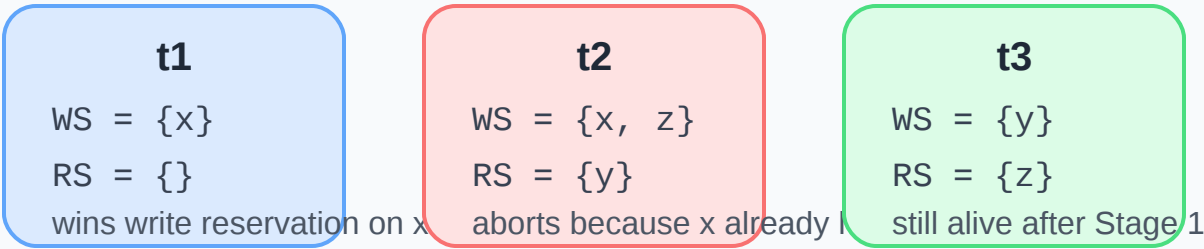


Conceptual Flow of AriaER

1 Stage 1: Write reservation and WAW detection

Transactions in the batch



Reservation table after write reservation

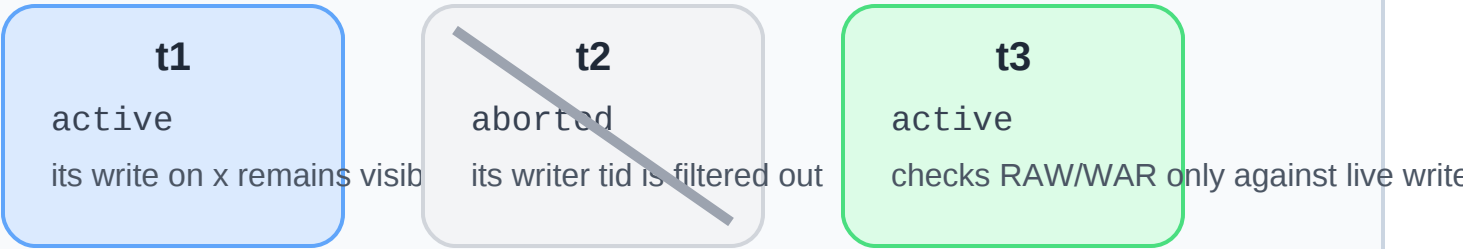
item	write_tid	interpretation
x	t1	t1 keeps the smallest tid
y	t3	t3 writes y
z	t2	written by t2 before WAW abort is known

Abort list after Stage 1



2 Stage 2: Skip WAW-aborted writers in RAW/WAR checking

Only surviving transactions continue



RAW/WAR check after abort filtering

item	writer tid	how Stage 2 treats it
x	t1	valid writer, used in dependency checks
z	t2	ignored because AbortList[t2] = 1

filtered out

t3 no longer observes the aborted writer t2
Result: t1 and t3 can commit in the same batch.

Key idea

AriaER performs WAW filtering first, then excludes WAW-aborted transactions from later read reservations and RAW/WAR dependency checks.